Permafrost Is Thawing From the Inside Out in Canada's North

From: Popular Mechanics Mined: 2026-11-07 5 min read

Deep in Canada's Northwest Territories, the ground itself is changing. **Permafrost**, ground that stays frozen for at least two years in a row, is thawing in surprising ways. In some places, it is melting from the inside out and setting off a new kind of landslide.

The Mackenzie River winds through this frozen landscape. It is North America's largest northward-flowing river and drains about 20 percent of Canada's land. For thousands of years, **permafrost** held the region steady. But as the climate warms, that stability is crumbling.

In 2017 and 2018, two huge landslides struck the central Mackenzie Valley, moving 4.5 and 18 million **cubic meters** of frozen ground. What puzzled scientists was what the landslides carried: large chunks of earth that were still frozen, which had slid hundreds of meters. Normally, **permafrost** melts from the top down into a slow muddy slide, but these slides were different. Ground observations showed the sliding happened about 20 meters below the surface.

In 2019, **permafrost geologist** Joseph Young, then a PhD student at the University of Alberta, flew to a remote field site and saw slopes collapsing with landslides everywhere. He and a team from the university tracked ground temperatures from 2020 to 2022. They concluded that wildfires from decades earlier and rising summer temperatures had formed cracks in the ground. Water **seeped** in through the cracks and warmed the **permafrost** from below, weakening its base. The researchers named the new phenomenon **permafrost** detachment slides **and reported it in the journal Geophysical Research Letters.**

Warming and wildfire together push heat underground over decades, weakening the **permafrost** base. Thawing also releases trapped metals into rivers, turning some orange, and studies estimate it could cause more than $50 billion in damage to buildings and roads in Alaska by 2050.

Word Treasure

permafrost -- Ground that stays frozen for at least two years in a row.

permafrost detachment slides -- A new kind of landslide in which frozen ground slides away from deep below the surface.

geologist -- A scientist who studies rocks and the ground.

seeped -- Slowly flowed or leaked through small cracks.

cubic meters -- A way of measuring how much space something fills.

Background you need

Permafrost is ground that stays frozen for at least two years in a row, and it acts like a solid base that holds northern landscapes steady.

For thousands of years, permafrost has held the region around the Mackenzie River steady, keeping slopes and soil in place.

When the climate warms, the frozen ground can thaw, and water seeping through cracks warmed the permafrost from below and weakened its base.

That weakening can set off landslides that move large chunks of still-frozen earth, and thawing can also release trapped metals into rivers and damage buildings and roads.

Break it down

LEAD: Permafrost in Canada's Northwest Territories is thawing in surprising ways, melting from the inside out and setting off a new kind of landslide.

KEY EVENT: In 2017 and 2018, two huge landslides struck the central Mackenzie Valley, moving 4.5 and 18 million cubic meters of frozen ground.

+- WHO: Permafrost geologist Joseph Young and a team from the University of Alberta

+- WHERE: The central Mackenzie Valley in Canada's Northwest Territories

+- WHEN: Landslides in 2017 and 2018; ground temperatures tracked from 2020 to 2022

EVIDENCE: The slides carried large chunks of earth that were still frozen and had slid hundreds of meters, and ground observations showed the sliding happened about 20 meters below the surface, unlike the usual slow muddy slide from the top down.

CAUSE: Wildfires from decades earlier and rising summer temperatures had formed cracks in the ground, and water seeped in through the cracks and warmed the permafrost from below, weakening its base.

OPEN QUESTION: How much damage thawing permafrost will cause, with studies estimating more than $50 billion in damage to buildings and roads in Alaska by 2050, and how it will change rivers as trapped metals are released.

Why it matters

Warming and wildfire together push heat underground over decades, weakening the permafrost that holds northern land steady. As it thaws, it can damage buildings and roads and release trapped metals into rivers, so understanding this new kind of landslide matters for the people and places that depend on frozen ground.

Test what you remember

1. What is permafrost?

- (A) Ground that freezes every winter and thaws every summer
- (B) Rock that has been frozen since the Ice Age
- (C) A thick layer of snow that never melts
- (D) Ground that stays frozen for at least two years in a row

2. Which river winds through this frozen landscape?

- (A) The Yukon River
- (B) The Mackenzie River
- (C) The Amazon River
- (D) The Mississippi River

3. How far below the surface did the sliding in the new kind of landslide happen?

- (A) About 20 meters below the surface
- (B) Just below the grass
- (C) About 200 meters below the surface
- (D) At the very top of the ground

4. What did the team conclude had formed cracks in the ground?

- (A) Heavy rain in a single storm
- (B) Roads built across the valley
- (C) Earthquakes under the river
- (D) Wildfires from decades earlier and rising summer temperatures

5. What name did the researchers give the new phenomenon?

- (A) Frozen river slides
- (B) Permafrost detachment slides
- (C) Deep ice cracks
- (D) Slow muddy slides

6. According to studies, what could thawing cause in Alaska by 2050?

- (A) Warmer winters for everyone
- (B) More than $50 billion in damage to buildings and roads
- (C) Fewer wildfires each year
- (D) New farmland across the north

Answer key (try the quiz first!): 1: D 2: B 3: A 4: D 5: B 6: B

Questions to think about

MY THOUGHTS (at least 20 words):